

Water as a Collaborative Medium: Rethinking Human-Nature Engagement

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Abstract

This PhD research investigates human-nature engagement through an HCI lens, focusing on water as both a medium and a metaphor for designing interactive systems. Employing a research-through-design approach, the study explores water's sensory, ecological, and symbolic properties by developing water-based technologies, such as ultrasonic water manipulation, to prototype scenarios of calm and meaningful interaction. Complementing this, ethnographic studies with coastal and urban water communities will examine human-water relationships, exploring how cultural practices and environmental challenges inform situated knowledge. The research aims to generate actionable design implications for technologies that foster meaningful human-nature engagement, with an emphasis on promoting sustainable practices and enhancing individual wellbeing.

CCS Concepts

• **Human-centered computing** → **Interaction design theory, concepts and paradigms**; *Ubiquitous computing*; *Interaction devices*.

Keywords

Human-nature engagement, water, sustainable wellbeing

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1 Motivation

It may seem like a distant fantasy, but what if, for future generations, nature exists only as a memory—reconstructed entirely in digital realms? Imagine the scent of wildflowers drifting through a spring meadow, the warmth of sunlight on skin, or the rhythmic buzz of insects on a summer breeze—all rendered by artificial intelligence and experienced only in Virtual Reality. As artificial environments become more sophisticated, this scenario may not be as far-fetched as it seems [20]. With over half of the world's population now living

in cities, urbanization, and technology have contributed to what Robert Pyle called the "extinction of experience"[25, 29]—the loss of meaningful connections with the natural world due to reduced exposure to nature. This disconnection not only diminishes wellbeing but also weakens future generations' motivation to address environmental issues.

As the concept of "Sustainable Wellbeing" [22] gains traction across disciplines such as conservation ecology, public health, and design research, the interconnectedness of human wellbeing and environmental sustainability has emerged as a critical focus. Addressing this multifaceted challenge requires more than abstract frameworks; it calls for practical, grounded strategies. As a PhD researcher in HCI, I investigate how interactive technologies—such as devices or installations—can foster sensory, emotional, and reflective engagements with nature. These engagements are designed to evoke moments of calm, mindfulness, and ecological awareness by incorporating natural elements like water or light. Drawing on metaphors of fluidity and resilience, my research aims to design contemplative experiences that encourage individuals to reconnect with their environment in meaningful and sustainable ways.

"I have no model for peace, only glimpses of it, metaphors for it, similes to what I cannot fully grasp and hold. Among them: a bowl of clear water. A boat drifting on a slow river. A lake among hills. The vast depths of the sea. A drop of water at the tip of a leaf. The sound of rain. The sound of a fountain. The bright dance of the water spray from a garden hose, the scent of wet earth."
— Ursula K. Le Guin

2 Background

According to Webber et al.'s review of HCI literature[32] on human-nature engagement, most research motivations fall into two categories: ecological aims promoting environmental sustainability, and humanistic aims focused on enhancing individual wellbeing. However, there's a growing recognition of the need for a socio-ecological systems approach that considers the mutual interdependence of humans and the environment. This aligns with the concept of "sustainable wellbeing"[22] which emphasizes using technology to reconnect people with the natural world.

Frauenberger's work[12] on entanglement in HCI explores how humans and technology shape one another. He emphasizes that technology should enhance, not distract from, human-nature engagement. One promising approach to awakening humans to the interconnectedness of the more-than-human world is through the act of noticing. Activities like hiking[1, 21], farming/gardening[17, 27], birdwatching[4, 18], foraging[16], or fermentation[2, 5] provide

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opportunities for daily engagement with other people and animals while fostering kinship through the mediation of technology.

While some focus on living beings, others include non-living entities in this entanglement[30]. They argue that human-made artifacts—whether a bowl[31], a loom[23], or artificial intelligence[6]—have a place in this interconnected “constituency”[30], expanding the scope of the investigation to include all things in the “Pluriverse.”[10] These ideas inspired me to explore the most ubiquitous and versatile entity—water—as a starting point for studying these relationships.

3 Research Objectives and Questions

Water connects humans to the world in a profound way. In nature, water offers a sensory richness and an inherent calm—whether in the rhythmic crash of waves, the gentle ripples of a still lake, or the fleeting beauty of a raindrop’s descent. Yet, how might we deepen and extend this connection through thoughtful design?

Posthuman Feminist Phenomenologist Astrida Neimanis[19] encourages us to think of ourselves as bodies of water and emphasizes that water is the element that binds us deeply to the environment. She describes this connection as the “hydrocommons” and recognizes that our bodies are an integral part of nature rather than separate or privileged. Building on this awareness, I strive to incorporate this understanding into the design, leveraging water as a material, metaphor, and more-than-human stakeholder to explore sustainable wellbeing.

RQ1: In what ways can the materialities of water, such as fluidity and transparency, be integrated into the design of interfaces?

With the proliferation of technologies aiming to blur the boundaries between the digital and physical material world [15, 24], there is a growing interest in the research and design of water-related experiences as evidenced by the expanding body of literature [7]. Water’s fluid, universal, and calming nature make it an ideal material for a tangible or ubiquitous interface. This question explores how water can be incorporated into a system to encourage users to engage with natural materials or phenomena. It examines the technical challenges (water sensing, water actuating, water energy/force) and opportunities of using water for interface design and how the sensory qualities of water—such as smell, sound, and movement—can be leveraged to create immersive, meaningful experiences that help connect people with the natural world.

RQ2: How do the aesthetic qualities of water influence user perceptions and emotions in interactive design?

This question explores how water’s aesthetic dimensions can influence emotional and sensory engagement in interaction design. It investigates how “somaesthetic qualities” (related to bodily awareness)[14], “fluid aesthetics” (focused on the dynamic and flowing nature of water)[11], and “eco aesthetics” (highlighting water’s connection to environmental awareness)[3] can be leveraged to create immersive experiences. The goal is to understand how these aesthetics can evoke emotional responses, promote mindfulness, and foster deeper connections to the natural world through interactive systems.

RQ3: What conceptual framework can water as a metaphor provide for critiquing and evaluating interaction design practices in technology?

This question explores how water’s metaphorical qualities—such as fluidity, adaptability, and interconnectedness—can shape design methodologies that emphasize the deep ties between human wellbeing and ecological responsibility. By integrating the principles of calm technology[33], which avoids distractions and enhances humans’ natural abilities, and embodied interaction[9], which focuses on delivering meaning through tangible and social computing, the research investigates how water can guide the creation of interactions that promote mindfulness and environmental awareness.

RQ4: How can water, understood as a more-than-human stakeholder, guide design solutions that consider its relationship with communities and ecosystems?

Viewing water as a more-than-human stakeholder means recognizing its agency and significance beyond human consumption. This involves paying close attention to the local communities whose livelihoods are intimately connected to water and gathering situated knowledge[13] through site-specific fieldwork. This question seeks to explore how design can address both human and environmental needs by treating water as an active participant in the process. It also aims to investigate how participatory design methods and technologies can be used to facilitate these interactions, fostering collaboration between people, water, and the environment.

4 Research Approach

My research employs research-through-design (RtD)[34] as its foundation, focusing on an iterative, design-led investigation into water’s role within HCI. As both a design researcher and practitioner, I create artifacts that engage with the materiality, fluidity, and symbolic resonance of water to foster sustainable and mindful user experiences. By developing tangible prototypes, I move beyond conceptual exploration, gaining hands-on insights into the possibilities and challenges of water as a dynamic design medium, and investigating how it can contribute to interactive, ecologically responsible systems.

This approach allows me to address technical challenges—such as actuating water using ultrasonic transducers and sensing through swept frequency capacitive sensing[28]—while exploring the conceptual aspects of water as an interactive, ecological, and metaphorical element. Through iterative engagement with these materials, I continuously refine the prototypes, revealing new design possibilities and tackling the complexities that arise from working with water as an unpredictable medium. This design process enables me to harness water’s potential as a dynamic material within HCI that promotes personal wellbeing and ecological responsibility.

My methodology interweaves field observations in blue spaces, hands-on prototyping, and theoretical reflection and this allows me to understand water’s aesthetic and experiential dimensions. By incorporating practices such as critical making[26] and Design Memoirs[8], I work not only to deepen my understanding of theoretical concepts—such as the “ways of water”—but also to explore their practical implications for human-water interactions. This approach underscores my commitment to investigating the intersection of theory, practice, design, and materiality by using water as a fundamental material in shaping human-environment interaction.



Fig. 1: Note from "Water Wednesday" field observation

5 Preliminary Results

Throughout my research, I have explored various aspects of water's potential in interaction design, particularly in relation to sustainable wellbeing. My ongoing "Water Wednesday"(Fig. 1) activity involves observing urban water bodies to acquire situated knowledge that will inform a field ethnography study on urban fishing scheduled from September 2024 to January 2025 (RQ4). Additionally, I have explored technical challenges surrounding water crest formation using ultrasonic transducers, conducting evaluations of water depth, container materials, and frequency impacts (RQ1)(Fig. 2). This work resulted in two technological configurations: continuous moving water pixels,(Fig. 3) or a transducer water pixel array (RQ1).(Fig. 4)

Building on these configurations, I developed several provocations(Fig 5) that each explore different water aesthetics—a water tank display, an interactive tea-tasting dish, a water plant terrarium, and a water light show—representing fluid, somatic, and eco aesthetics (RQ2). These prototypes not only demonstrate water's diverse sensory potential but also inspire reflection on its environmental significance. My conceptual contributions are reflected in the new framework for water-inspired technology design that was presented in my paper, "The Ways of Water: A Guiding Metaphor for Designing Calm Technology," published at the Halfway to the Future 2024 conference (RQ3).

During my internship at NTT Social Informatics Laboratories, I conducted preliminary research on sustainable wellbeing, culminating in plans for field research exploring water's relationship with rural communities in Japan. This work will further investigate how design and technology can promote more meaningful human-water interactions in everyday life.

6 Expected Contribution

The contributions of my research aim to explore several areas with a practical and grounded approach. First, it investigates how natural elements like water can be used in interactive systems, such as through ultrasonic manipulation, highlighting water's sensory and ecological potential. Additionally, through prototyping with water, I hope to gather insights and draw guidelines for creating calm, non-intrusive technologies. Lastly, my research aspires to engage with communities and ecosystems while emphasizing co-design and in

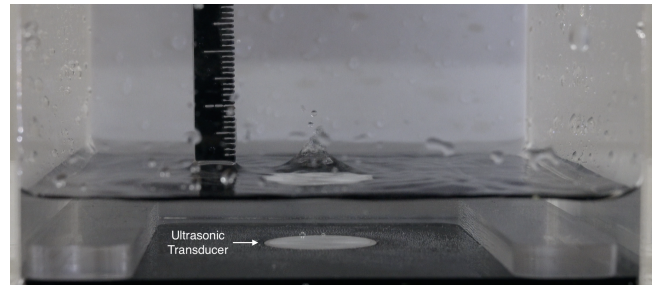


Figure 2: water crest created by ultrasonic transducer

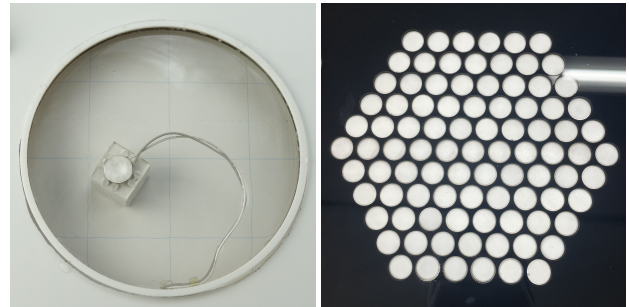


Figure 3: moving water pixel Figure 4: water pixel array mechanism with toio robot

situ work to learn from real-world applications and contribute to sustainable wellbeing.

7 Timeline

I am currently in the second year of my PhD program. During my first year, I participated in the cross-university Interactive Fluid Interface Project, funded by JSPS KAKENHI, where I explored water manipulation techniques using ultrasonic transducers (RQ2). My prototypes were exhibited at multiple venues, including the Ichikawa Art Festival, Lapin et Halot Gallery, International House Japan, and the 2024 Ars Electronica Festival(Fig 6). This exploration sparked deeper reflections on water's design potential and led to the development of a theoretical framework for calm technology inspired by water, presented in my paper, "The Ways of Water: A Guiding Metaphor for Designing Calm Technology," at the 2024 Halfway to the Future Conference (RQ3). Following this, I have proposed workshops on design ideation with water, planned for mid-2025.

In early 2023, I began Water Wednesday, a practice focused on observing natural water environments, to investigate human-water relationships (RQ4). Additionally, I am conducting an ethnographic study with Japan's recreational fishing community, which will run from late 2024 to early 2025. Starting at the end of 2024, I will also join the Social Cooperation Program between NTT R&D and the University of Tokyo on Sustainable Wellbeing to explore how water-centric designs can benefit local communities, especially the elderly (RQ4).

I plan to submit my dissertation proposal by mid-2025 and aim to defend it by Spring 2026.



Figure 5: “Water Containers” provocations: a) Water Tank Display b) Interactive Dish for Tea Tasting c) Water Plant Terrarium.



Figure 6: Signs of Water in Ars Electronica ©vog.photo cropped from original

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